BRASS containing the STRUPLO program can be downloaded from the *BRASS* homepage (http://www.brass.uni-bremen.de/) of the Central Laboratory for Crystallography and Applied Material Sciences (ZEKAM) at the university of Bremen. After download, an install routine will guide through the installation procedure.

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PDFgetX2: a user-friendly program to obtain the pair distribution function from X-ray powder diffraction data

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1. Overview

The program PDFgetX2 [1] processes X-ray powder diffraction data to obtain the pair distribution function (PDF). The PDF method directly measures the inter-atomic distances in the intuitive real space [2]. For a long time, this has been the method of choice for studying the structure of amorphous materials and liquids [3]. Recent advances in the PDF technique have shown it to be a powerful local structure probe of nano-crystalline and highly crystalline materials [4]. One key to a proper PDF analysis is to quantitatively extract both Bragg peaks (if any) and the diffuse scattering intensities containing the average and local structural information, respectively. PDFgetX2 aims to facilitate the user interaction with data from laboratory or synchrotron X-ray measurements during this otherwise arduous and intensive process. This program has the following features. i) User-friendly interface. All user inputs are through series of self-explanatory graphic user interfaces (GUI). User inputs are also kept to the minimum with built-in commonly used settings and atomic X-ray scattering data look-up tables. ii) Flexible data corrections. A rather large number of corrections are available, and each of them can be turned on/off independently using the GUI. iii) Easy problem diagnosis. Instead of being a black box, PDFgetX2 makes every intermediate correction term and its effect on the data accessible by a simple button click. iv) Complete history file for time saving and reproducibility. All data processing parameters are saved together with the data, and can be reloaded back to fully reproduce an earlier analysis. v) Efficient use and documentation. A simple parameter optimization algorithm is available to fine tune the PDF quality. The program executable and a user's guide are downloadable from PDFgetX2 web page at http://www.pa.msu.edu/cmp/billinge-group/programs/PDFgetX2/ and from the CCP14 website. A short tutorial with example data sets can be found in the user's guide.

2. Program Description

The program PDFgetX2 is written in IDL[1]. It runs on the freely downloadable IDL Virtual Machine (IDL[VM]) (available at http://www.rsinc.com/download/), which is supported on Linux/UNIX, WINDOWS, and MACINTOSH platforms.

Key implemented corrections in PDFgetX2 are based on the formula of the measured scattering intensity I_{raw} (each term is described below):

$$I_{raw} = NMAP(I_{coh} + I_{comp}) + I_{bkg} \qquad (1)$$

The goal is to recover the coherent intensity I_{coh}, as the only term conveying atomic structure information. The PDF, $G(r)$, is obtained by the sine Fourier transform,

$$G(r) = \frac{2}{\pi}\int_0^\infty Q[S(Q)-1]\sin(Qr)dQ, \qquad (2)$$

where $S(Q) = \frac{I_{coh} - < f^2 >}{< f >^2} + 1$ and Q is the magnitude of the momentum transfer. The commonly applied corrections, described in detail in [2], are:

1. Background scattering I_{bkg} subtraction. The I_{bkg} should be measured under the sample experimental condition without the sample.
2. X-ray polarization correction P. The X-ray polarization factor is the only parameter.
3. Sample self-absorption correction A. Parameters used here are the measured sample absorption coefficient and the sample geometry (transmission, reflection, etc.).
4. Multiple scattering correction M. This term is only significant in the case of strongly absorbing samples.
5. Compton intensity I_{comp} subtraction. The sample composition is needed to compute the theoretical Compton scattering profile. If the Compton contributed at high Q is removed by energy discrimination, a window function can be applied [5].
6. Normalization factor N. The high Q method is used for automatic normalization. Theoretical elastic scattering intensities are computed given the sample composition.

Additional corrections may be necessitated depending on specific experimental setups. Particularly, for the recent RA-PDF development that makes use of 2D detectors [6], oblique incident angle correction [7] and empirical energy dependence of the detection efficiency [8] are available in the program. An optional constant fluorescence intensity subtraction is also implemented.

Standard uncertainties due to finite counting statistics are estimated (if not present in the input data) and propagated in all steps. The final *S(Q)* and *G(r)* data files (automatically saved by default) are multiple-column ASCII files with the processing parameters in the header. The *S(Q)* data also contain the Faber-Ziman coefficients for all partial structure factors as additional columns. The *G(r)* file format is compatible with the PDF modeling programs PDFFIT [9] and DISCUS [10].

3. A Quick Example

To get some practical sense about the use of PDFgetX2, we will quickly go through one RA-PDF data processing example from a new binary antimonide, Ti_2Sb (see Ref. [11] for more detailed information). The goal of this PDF study was to resolve the local Ti square net distortions hinted by crystallographic analysis. The powder sample (around 0.5 g in flat plate transmission geometry) was measured at the high energy station 6-ID-D at the Advance Photon Source (APS) at Argonne National Laboratory. A short data collection time of 40 seconds was achieved with the RA-PDF method employing an image plate area detector. The raw 2D data were first integrated using the program FIT2D [12] to get the two column data $Q/2\theta$ vs. I. Then we are ready to fire up PDFgetX2 to obtain the PDF. Fig. 1 gives a glimpse of the program GUI.

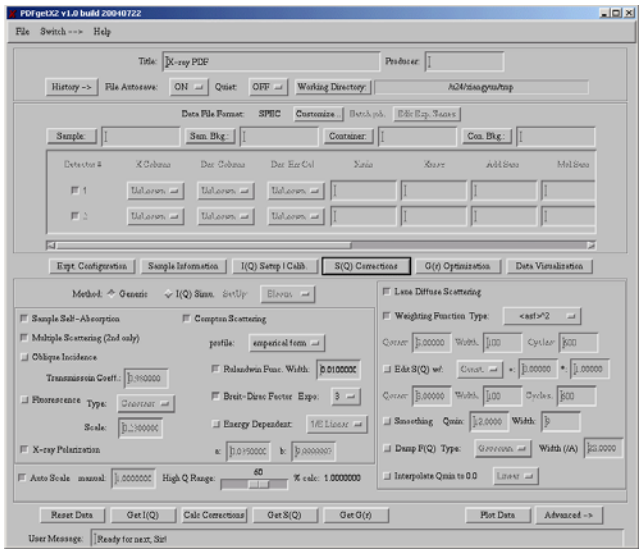


Figure 1: *A glimpse of the program PDFgetX2 GUI.*

The first phase of data analysis is to tell the program about the experiment (all with GUIs). This can be divided into four categories (the parameter values for this example will be shown in following parentheses). i) The scattering intensity profiles. Here we select the sample and background data files by clicking on the corresponding buttons. Additional information, such as the data file format (CHI) and the type of the data X column (2θ) should also be specified. ii) The X-ray properties. This includes the X-ray wavelength used (0.1266 Å) and the polarization factor (1.0). iii) Sample information. Elements are selected from the periodic table graph; atomic fractions are typed into the corresponding fields (Ti:2, Sb:1). The sample geometry (flat plate IP transmission) and the measured sample absorption coefficient (0.50) are also required. iv) The set of corrections to apply and some additional correction specific parameters if necessary. In addition to the default corrections, this RA-PDF setup requires at least the sample absorption, oblique incidence, and Compton scattering corrections. We shall start with these selections.

The second phase is to crunch the data and improve the PDF quality. Fortunately, only category (iv) needs to be adjusted during the second phase (maybe except when the range of data used is to be changed). The starting point is the scattering intensity $I_{raw}(Q)$ as shown in Fig. 2(a). It is generally straightforward to obtain a high quality PDF when the required processing parameters are determined experimentally. However, we would find the above settings result in a rather poor quality PDF, e.g., as shown in Fig. 2(b). This is because the RA-PDF method imposes further implications due to the lack of energy resolution of the image plate detector. There are two main issues: the fluorescence intensity and the energy dependence of the detection efficiency. So we switch these two corrections on. However, these two corrections are not convenient to determine experimentally. Here we can either choose to employ the trial-error approach or take advantage of the optimization feature of the program to find the optimal parameter values. In either case, a high quality PDF as shown in Fig. 2(c) (blue solid circle) results.

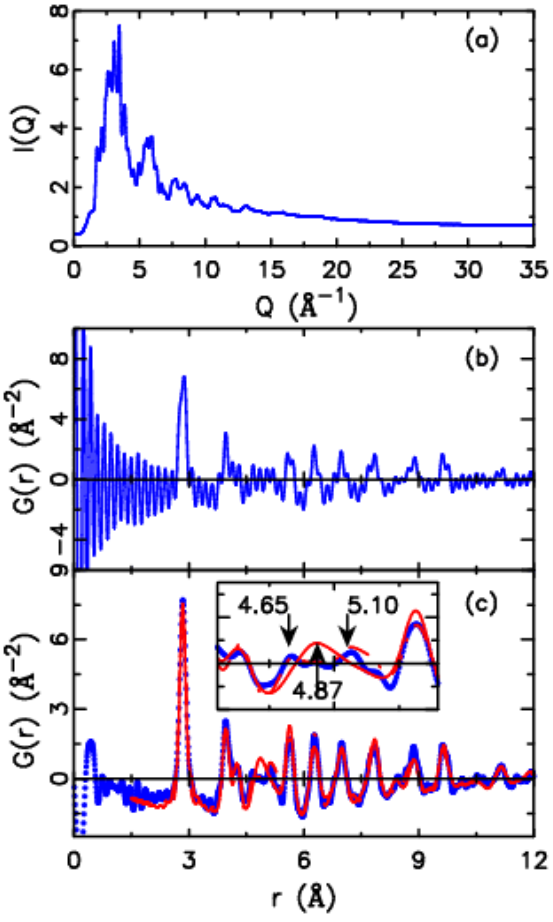


Figure 2: *(a) The scattering intensity I_{raw} of Ti_2Sb vs. Q. (b) One poor quality PDF resulting from insufficient corrections. (c) The experimental PDF after proper corrections (blue solid circle), the simulated PDF from the distorted structure model (red dashed line), and the simulated PDF from the undistorted model (red solid line). Inset zooms in the 4-5 Å region.*

Shown also in Fig. 2(c) are the model PDFs calculated from the undistorted Ti_2Sb crystal structure (red solid line) and the distorted structure (red dashed line). The most

pronounced evidence of the superiority of the distorted model appears in the 4-5 Å region. The single peak (4.87 Å) in the undistorted model splits into a doublet (4.65 and 5.10 Å) which is clearly in the data and also the distorted model. The experimental PDF provides direct proof of the existence of local Ti structure distortions.

Finally, the program PDFgetX2 can work without the GUIs through manual function calls. This requires knowledge of IDL and internal PDFgetX2 structure to some level. This makes it possible for PDFgetX2 to batch process multiple data sets and interface with other programs.

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BRASS, the Bremen Rietveld Analysis and Structure Suite

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BRASS [1,2] is a common platform for a suite of programs covering the display and Rietveld analysis of powder diffraction data and closely related tasks such as Fourier and grid search calculations and the drawing and crystal chemical evaluation of structure models.

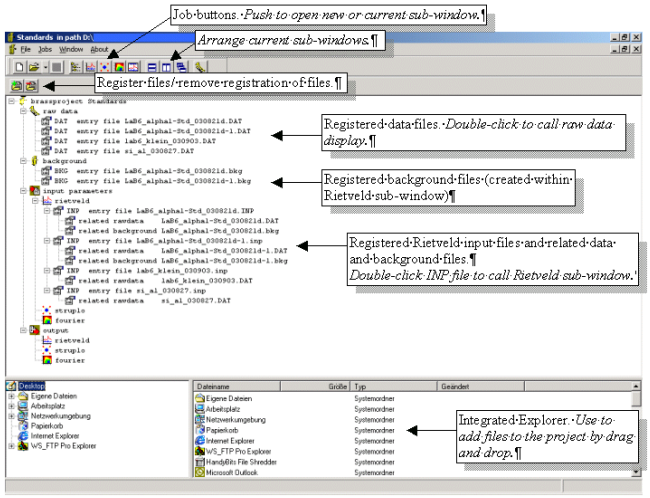
Currently accepted powder diffraction data formats are (contact us for further types to be incorporated):

- DAT = PC-Rietveld+ format
- RAW = Siemens binary data
- RD = Philips binary data
- UDF = Philips ASCII data
- XRDML = PANalytical html-type ASCII data

The general organization of BRASS:

- Main window contains all different types of jobs (Rietveld, Struplo, Fourier, Pattern display) as sub-windows
- Project manager sub-window (see below) provides access to registration and unregistering of files by drag and drop (or add/ remove buttons) and to the call of the different jobs (Rietveld, Struplo, Fourier, Pattern display) by double click.
- Job sub-windows may be opened in parallel.

Each job type runs separately but information is easily exchanged by dedicated buttons.



Rietveld features:

- Comprehensive graphical user interface.
- The background may be defined by a polynomial function or graphically by set points for linear interpolation
- Separation of instrumental and sample effects based on the reformulation of the Pseudo Voigt function according to [2, 3]